
HALO REACH: Modular Add-ons Mechanism Design and Analysis

UAS for Disaster Relief - AE3 2024



Student: Andrew Nakhla

CID: 02044646

Supervisor: Dr Masoud Seifikar

Department: Department of Aeronautics

Course: MEng Aeronautical Engineering

Module: AERO60004 - Group Design Project

Date: 17/06/2024

Academic Year: 2023-2024

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.



Table 1. People involved in HALO REACH - GDP

Team code	Team	Name	CID
FATT01	Team Leads	Moreno, Patricia	01857449
		Rashid, Hasan H	02020313
FATT02	Aerodynamics	Alam, Shahriar	02037395
		Dirirsa, Naboth	02026211
		Lawlor, Max F	02026199
		Singh, Parasdeep	02099678
		Homa, Matthew M	02038403
FATT03	Structures	Leng, Jiaxun (Aaron)	02015060
		Wang, Zhiyan	01911209
		Liu, Stephen	02015258
		Chauhan, Shivani	02096006
FATT04	Systems and propulsion	Lin, Yongkai	02029838
		Chursuk, Tharm	02048996
		Wankhade, Divij P	02027906
		Aggarwal, Ishaan K	02030969
		Halim, George	02059571
FATT05	Flight dynamics	Baikady, Thilak S	02016794
		Scott-Brown, Ruairidh	01734080
		Allen, Harry	02028116
		Tang, Shun S	02023280
		Lee, Jeongheon (Edward)	01754474
FATT06	Add-ons	Rosenthal, Adam T	02027024
		Ramteke, Shreya	02079050
		Nakhla, Andrew	02044646

Abstract

The Modular add-ons is a crucial part of this VTOL UAS which is designed for disaster relief operations. This report dives into the design process and decisions made to find the best and most efficient mechanism for the modular add-ons to be swapped, contributing to the success of two different missions. The report will detail the attachment methods, material selection, manufacturing methods, aerodynamic analysis and how this UAS can also be used for other markets. It will also focus on why a mechanical clamp was chosen and why the material Epoxy carbon fiber was used. Calculations in the report will also show how minimal the extra drag coefficient (0.00732) is due to the modular add-ons. A more in-depth explanation of how this UAS will be used for agriculture and monitoring will also be given in this report.

Contents

Table of Contents	iii
List of Figures	iv
List of Tables	v
Nomenclature	vi
1 Introduction	1
2 UAV Configuration	1
3 Modular Add-ons Attachment Method	1
3.1 Sliding Rails	2
3.2 Latch and Lock system	2
3.3 Electromagnets	2
3.4 Spring Loaded Pins	2
3.5 Mechanical Clamps	3
3.6 Final Choice	3
4 Clamp Bottom Plate	4
4.1 Material Selection	4
4.2 Manufacturing Method	6
4.2.1 UD Prepreg	6
4.2.2 UD Lay-Up	6
5 Attachment of Clamp to Fuselage	7
5.1 Rod, Bearings and Motor	7
5.2 Stringers and Spring Reels	8
6 Aerodynamic Analysis	9
6.1 Miscellaneous Drag	10
6.2 Leakage Drag	10
7 Communication Relay	10
7.1 Payload Container	10
7.2 Bottom Plate	12
8 Other Market Uses	12
9 Conclusion	12
References	13
A Clamp's Bottom Plate Covering Slot	14
B Heavy Frames	15
C Final Clamp Attachment To Fuselage Design	16
D Final Clamp's Bottom Plate Design	17
E Communication Relay Bottom Plate	18

List of Figures

1	Two different configurations to store the add-ons payload	2
2	Clamp with payload	4
3	Ashby chart for the clamp's bottom plate	5
4	Clamp's rod, rails, gears and motor	7
5	Structural analysis of bottom plate	8
6	Final clamp attachment to fuselage design with stringers [1]	9
7	View of underside of fuselage with clamp and add-ons	9
8	Top side view of the new payload container with opening for wiring	11
9	3D printed structure to hold router (in orange)	11
10	Bottom plate covering slot	14
11	Image to show heavy frames on a fuselage	15
12	Structural analysis of final clamp attachment to fuselage design with stringers [1] . . .	16
13	Final clamp's bottom plate design [1]	17
14	Communication relay bottom plate with antenna opening	18

List of Tables

1	People involved in HALO REACH - GDP	i
2	Pugh Matrix for Attachment Methods	3
3	Pugh Matrix for Attachment Methods continued	3
4	Clamp bottom plate's material properties	6

Nomenclature

ρ	Density (kg/m ³)
σ_{tensile}	Tensile Strength (MPa)
σ_{10^7}	Fatigue Strength at 10 ⁷ cycles (MPa)
A_{frontal}	Frontal Area (m ²)
c	Cost (GBP/kg)
C_D	Drag Coefficient
E	Young's Modulus (GPa)
S_{ref}	Wing Reference Area (m ²)
GTOW	Gross Takeoff Weight
HALO	Humanitarian Aid Logistics Operations
M	Merit Index
REACH	Rapid Emergency Aid and Communications Hub
UAS	Uncrewed Aerial System
UAV	Unmanned Aerial Vehicle
VTOL	Vertical Take-Off and Landing

1 Introduction

Following the recent earthquakes in Syria and Turkey, the importance of VTOL UAS for use in disaster relief missions have increased, as traditional ground and aerial vehicles often struggle to access the affected areas due to infrastructure damage and severe weather conditions. The UAS for Disaster Relief (UDR24) project aims to overcome these challenges by designing a modular VTOL UAS capable of vertical take-offs and landings from ship decks, even under adverse conditions. The UAS should serve as a long-endurance communications relay or land vertically to deliver relief supplies. The main objectives for the modular add-ons team is:

- To design and implement modular components that allow the UAS to switch between the two different mission profiles.
- To ensure the modular add-ons can be swapped in and out within 10 minutes by two people between the missions.

The objectives for this report is to: decide on the most suitable attachment method for the add-ons, select an appropriate material and manufacturing method for the bottom plate, design how the clamp will be attached to the fuselage, study the aerodynamic effects of the add-ons, design the long-endurance mission (communication relay) payload container and bottom plate, and finally, determine how this UAV can be used for other markets.

2 UAV Configuration

To initiate the design process, selecting an optimal configuration for the UAV was important. A comprehensive meeting involving all subgroups was held to evaluate various UAV configurations. The configurations under consideration included vectored thrust, tilt wing, wingless, VTOL fixed wing, MALE/HALE, quadcopter, conventional single rotor, coaxial single rotor, and hexacopter. Each subgroup assessed these configurations based on how well they supported their specific design requirements.

The ratings from each subgroup were then aggregated to identify the most suitable configuration for the project. The VTOL fixed wing configuration emerged as the best choice. This configuration has multiple advantages for the modular add-ons team, it does not only facilitate easy payload access and swapping, but it is also well-suited for the long-endurance mission profile.

3 Modular Add-ons Attachment Method

As the payload for the short-endurance mission (delivering relief supplies) is 50 kg, and since there is a restriction on the payload being swapped in less than 10 minutes, the attachment method of the modular add-on to the UAV had to be chosen wisely. Research was done to try and find similar aircrafts with modular add-ons but none were found as the requirements for this UAV are different. All modular add-ons team members then worked together to come up with ideas and design how the add-ons will be attached and swapped.

Before deciding on the attachment method, it was necessary to first determine approximately where in the fuselage the add-ons will be stored since Systems and Propulsion (weights and balance) and Structures teams are the ones responsible of the exact position of the add-ons along the fuselage on its underside (as underside is easier for user to load/unload payload than other locations due to the configuration of the UAV which includes the propellers and height of UAV). Figure 1 below shows two different configurations to store the add-ons, it was decided to go with the 'slot' configuration (option 2) for multiple reasons, these include that it is more aerodynamic efficient as the shape of the fuselage is maintained (no bluff body as in other option), the payload is kept protected as it is covered by all sides and the payload is also more secured in place when compared to the other option.

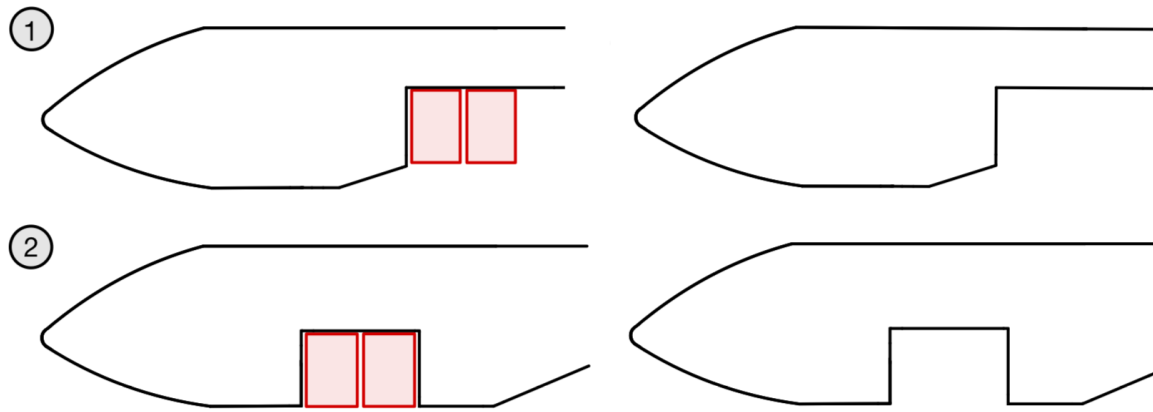


Figure 1. Two different configurations to store the add-ons payload

After deciding on the slot configuration, the next step was then to determine the optimal method for attaching the add-ons to the UAV, various approaches were considered. These methods were thoroughly studied and evaluated by all team members, as detailed below.

3.1 Sliding Rails

One method considered was using sliding rails attached to the fuselage, which would allow modular add-on containers to slide and be secured onto those rails. This attachment method is straightforward and easy to use. However, the complexity arises when multiple payload containers are involved, as accessing a specific container requires unloading all the containers in front of it first. Additionally, devising a way to lock the containers in place added to the design's complexity. Other advantages included that this method required no power to operate and it also allowed the containers to be moved along the fuselage to help with balancing.

3.2 Latch and Lock system

The latch and lock system works by having multiple chains which are attached to the payload container, these chains would then latch into hooks on the fuselage and these hooks would then lock the chains in place. Although this method is simple and requires no power usage, it does have some disadvantages when used for this UAV. The latch and lock mechanism would take more time to use when compared to other methods as each chain would have to be latched and secured in place manually and this would therefore stop the swapping in under 10 minutes objective from being achieved. The reliability of this method is also not the best, as the latching mechanism (hooks) may fail and stop the user from locking/unlocking the chains and so the mission would fail.

3.3 Electromagnets

Using electromagnets to attach the modular add-ons to the fuselage is a very simple design, multiple electromagnets will be attached to the fuselage and connected to power, these will then attract the payload container which will at least have the top plate made out of a magnetic metal. This method is easy and fairly simple to use, but it will require constant power consumption to turn the electromagnets on and keep the payload held in place. Another fear of using this method is that the magnetic field from the electromagnets might interfere with the communication system of the UAS and this would therefore make the UAV uncontrollable.

3.4 Spring Loaded Pins

Another method was to use spring loaded pins which would allow the payload container to be held in place when the pins are compressed and locked, it can then be easily released to allow swapping or

removing the container. The primary advantage of this approach is its simplicity and speed, as the pins can quickly lock the containers into place with minimal effort. Additionally, this method does not require any power consumption, making it highly efficient. However the main challenge is ensuring the pins provide a secure attachment under all flight conditions, the reliability of the spring mechanism over multiple uses also need to be carefully considered to prevent any potential failures during critical missions.

3.5 Mechanical Clamps

Mechanical clamps were considered as a method for attaching the modular add-ons to the UAV. This method involves using robust clamps to secure the payload containers within the designated slot, similar to how a forklift lowers and raises a load. The reliability of the clamps and the secure attachment they provide make it well suited for the UAV. They also do not require any power usage. However, there will be more manual effort required from the user to wind and lower/raise the payload. As the clamp will have a plate on which the payload will be placed, this is advantageous aerodynamically as when the UAV is carrying no payload, then the empty slot will be covered by the plate and prevent any turbulence or separation that that could be caused by the open slot. Figure 10 in appendix A below shows how the plate covers the slot.

3.6 Final Choice

After research and much discussion about each attachment method, a Pugh matrix was created and filled together with the rest of the modular add-ons team. The characteristics considered can be seen in the Pugh matrix below in tables 2 and 3, as it can be seen, the 'ease of use' was the most important characteristic as one of the main objectives of this project was that the modular add-ons must be swappable in under 10 minutes.

Table 2. Pugh Matrix for Attachment Methods

Characteristics	Weightings	Sliding rails	Latch & lock	Electromagnets
Ease of Use	10	4	3	5
Structural Compatibility (incl Weight)	8	3	3	2
Simplicity	7	4	4	5
Power Consumption	5	5	5	2
Versatility/Modularity	8	4	2	5
Reliability	7	5	3	3
Total		184	144	172

Table 3. Pugh Matrix for Attachment Methods continued

Characteristics	Weightings	Spring loaded pins	Mechanical clamps
Ease of Use	10	4	2
Structural Compatibility (incl Weight)	8	3	3
Simplicity	7	5	4
Power Consumption	5	5	5
Versatility/Modularity	8	2	4
Reliability	7	3	5
Total		161	164

The top three choices were then selected (it can be seen from the tables that these are the sliding rails, electromagnets and mechanical clamps), each of these choices were then carefully analysed and further research was done on them to help choose the best option for the UAV. Some simple structural analysis was also done on these three different attachment methods by [1]. The total weight of the whole modular add-ons swapping system (payload container + attachment mechanism) was restricted by the brief [2], and the Flight dynamics and Structures teams. After conducting the structural analysis and reviewing the advantages and disadvantages of each method for use with this UAV, the mechanical clamp mechanism was chosen. This method was deemed the most suitable considering factors such as power consumption, reliability, required effort, and safety, while also providing excellent structural analysis results. The mechanical clamps were also the only mechanism that allowed the payload to be lowered/raised without having to design another mechanism to do so and this was one of the main reasons the clamps were chosen. Mechanical clamps are also robust and provide a secure attachment, ensuring that the payload remains stable during flight. Additionally, the ease of use and quick release mechanism of mechanical clamps allow for the payload to be swapped within the required 10 minute timeframe, to also help achieve this requirement, a small efficient motor can be used to lower and raise the payload, reducing the manual effort required from the user, with negligible power consumption. This method also maintains the aerodynamic efficiency of the UAV, as the bottom plate can be seamlessly integrated into the fuselage design. Figure 2 below shows how the clamp will work and carry the payload (modular add-ons), the gears and rails will be facing the back-end and therefore allow the payload to slide towards the front or sideways. The density and volume of the payload container were given by the brief [2], it was also mentioned that the container had to be a cube, therefore after simple calculations, the side length of the container was found to be 36.9 cm. The Structures team was also informed of these dimensions to design the slot and ensure there is the required ground clearance for loading/unloading the payload, as they were designing the landing gear.

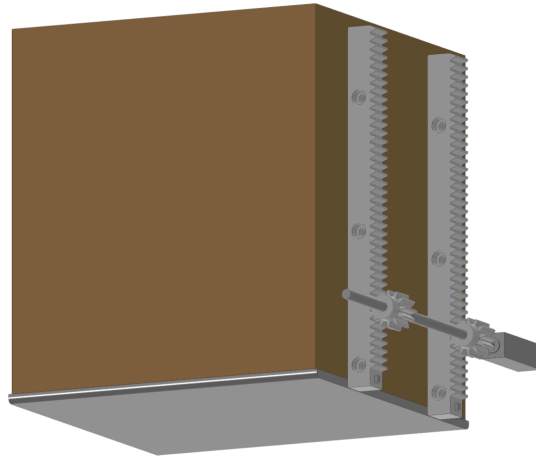


Figure 2. Clamp with payload

4 Clamp Bottom Plate

4.1 Material Selection

Selecting the appropriate material for the clamp's bottom plate is crucial to ensure the secure attachment and overall performance of the clamp. Various materials were considered, taking into account the following factors:

- **Strength and Durability:** The plate must withstand the forces experienced during flight and the repeated loading and unloading of the payloads without degrading over time.
- **Weight:** Minimising weight is essential to maximize the UAV's range and fuel efficiency. This is

also an important factor as the total weight of the add-ons mechanism is restricted to a maximum of 15% of the GTOW by the brief [2], however this was then further restricted to a maximum of 7 kg by the Flight dynamics team.

- **Cost and Availability:** Practical considerations such as the cost of the material and its availability also played a significant role in the selection process, with an emphasis on minimising costs while ensuring the material met all necessary performance criteria.

One method which can help with the material selection process is to use a merit index equation, the main idea is that an equation for the merit index M is derived and the main purpose is to maximise it. Equation (1) below shows the merit index equation where σ_{tensile} is the tensile strength, σ_{10^7} is the fatigue strength at 10^7 cycles, ρ is the density and c is the cost. The raising of the tensile strength to the power of $1/2$ comes from theory [3] where the merit index is derived for a plate/panel loaded in the same way as this plate with the clamp. σ_{10^7} , ρ and c were then added to consider the factors which are mentioned above.

$$M = \frac{\sigma_{\text{tensile}}^{1/2} \sigma_{10^7}}{\rho c} \quad (1)$$

Since the main idea was to maximise tensile strength and fatigue strength (since the multiple loading and unloading cycles would open up cracks) and minimise density and cost, it can be seen from the equation that this is equivalent to maximising M , as the tensile and fatigue strengths are in the numerator while the density and cost are in the denominator.

Using the merit index above, to help choose a material for the bottom plate of the clamp, the CES Granta software [4] was then used to plot a graph with all the materials in a graph with $\sigma_{\text{tensile}}^{1/2} \times \sigma_{10^7}$ in the y-axis and $\rho \times c$ in the x-axis. A line with gradient 1 was then plotted and moved upwards to help select the highest material in the top left hand corner as this is the material with the highest M (numerator maximised and denominator minimised). Figure 3 below shows the Ashby chart for the material selection, it can be seen that Epoxy/HS carbon fiber is the material with the highest value for M , and it was therefore chosen to manufacture the bottom plate of the clamp.

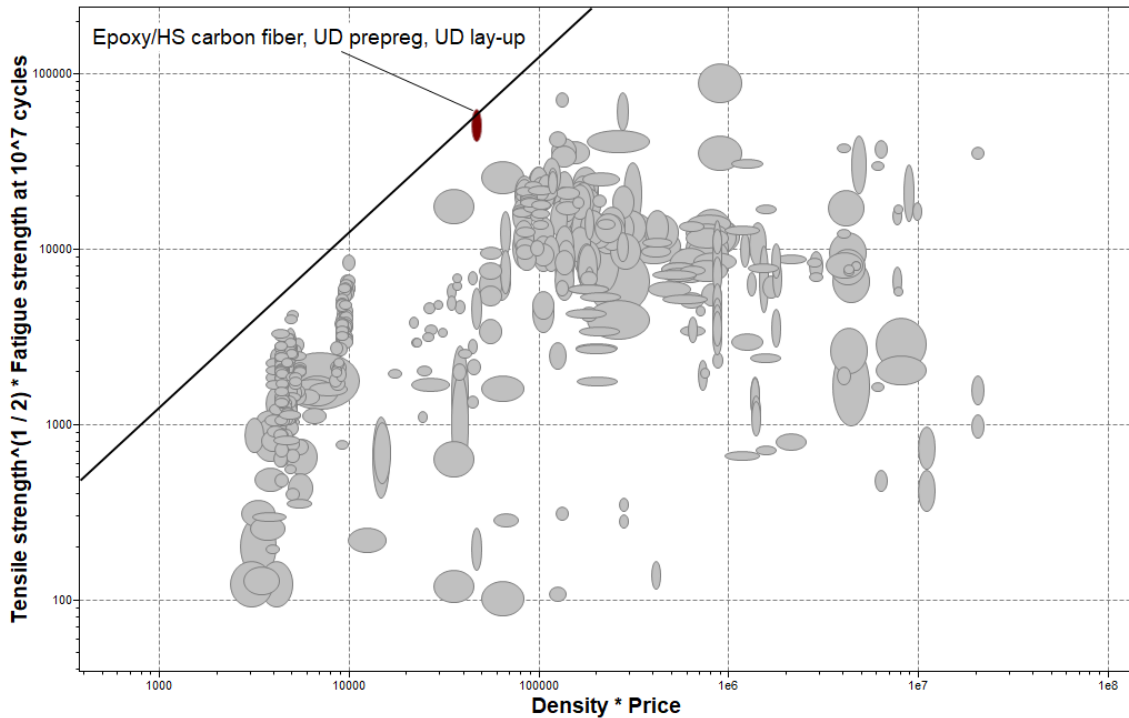


Figure 3. Ashby chart for the clamp's bottom plate

Table 4 below shows the material chosen for the bottom plate and its properties.

Table 4. Clamp bottom plate's material properties

Material	ρ (Kg/m ³)	σ_{tensile} (MPa)	σ_{107} (MPa)	E (GPa)	cost (GBP/Kg)
Epoxy/HS carbon fiber	1.55×10^3	2.17×10^3	1.41×10^3	154	28.10

4.2 Manufacturing Method

After choosing the material, the manufacturing method for the plate had to be chosen, and using Epoxy/HS carbon fiber involves several steps that ensure the structural integrity and performance of the component. Below are the details for the processes of using UD Prepreg and UD Lay-Up techniques to manufacture the plate.

4.2.1 UD Prepreg

UD (unidirectional) prepreg refers to pre-infused carbon fiber sheets where the resin system (epoxy) is already integrated into the fiber. This method offers high-quality and consistent properties.

Process steps:

1. **Preparation:** Prepare a mould (positive and negative moulds) and add releasing agent to ensure easy removal of the cured plate.
2. **Lay-Up:** Cut the UD prepreg sheets to the required dimensions, then lay the sheets in the desired orientation on the mould. This often involves multiple layers stacked to achieve the required thickness, ensuring proper alignment of fibers to optimise strength and stiffness.
3. **Vacuum Bagging:** Cover the lay-up with a vacuum bag, then seal the bag and apply a vacuum to remove any trapped air and compact the layers together.
4. **Curing:** Place the vacuum-bagged lay-up in an oven and cure, this step ensures the resin fully polymerises, bonding the fibers into a solid plate.
5. **Cooling and Demoulding:** After curing, allow the plate to cool down gradually to room temperature before removing it from the mould.
6. **Trimming and Finishing:** Trim the edges of the plate to the final dimensions and perform any necessary finishing operations, such as sanding.

4.2.2 UD Lay-Up

UD Lay-Up involves placing layers of unidirectional carbon fiber sheets and manually infusing them with epoxy resin. This method is more labor intensive but offers flexibility in the manufacturing process. The process steps are very similar to the UD prepreg, but the difference is that after placing each carbon fiber sheet, one would need to apply the epoxy resin uniformly over the layer using a brush or roller. There is also another extra step which involves using rollers or squeegees to compact the layers and remove any trapped air bubbles, this step is critical to ensure proper fiber-epoxy distribution and to avoid voids.

After carefully comparing the two methods, it was decided to go with the UD prepreg method for manufacturing the plate for several reasons. The UD prepreg method ensures more consistent quality and produces a product with higher mechanical properties and fewer voids due to the uniform epoxy distribution when compared to the lay-up method. The efficiency is also better with the prepreg method, as there is reduced manual effort unlike the lay-up method, which requires epoxy to be manually applied after each layer. Although the initial costs are higher because prepreg materials are

often more expensive, these costs are justified by the critical role of the bottom plate in the overall performance and safety of the UAV.

5 Attachment of Clamp to Fuselage

The clamp will be carrying payloads of up to 50 kg, and the total weight of the clamp itself will be around 7 kg. Therefore, it is crucial to consider these weights when designing the attachment method for securing the clamp to the fuselage. The following subsections explain how the clamp is attached to the fuselage and held in place.

5.1 Rod, Bearings and Motor

The design process of how the clamp will raise and lower the payload, along with the sizing and calculations of the mechanism, can be found in [5]. The clamp operates using gears and rails, the gears will be connected together using a rod to a motor which will be able to spin them to lower or raise the bottom plate.

Figure 4 below shows the gears, rails, rod and motor. It can be seen from the figure that the clamp can only be supported from the motor and rod. Therefore, after providing the Structures team with the total weight of the clamp and the heavy payload (around 57 kg), some calculations were conducted by both teams and it was then agreed to go with the idea of using a heavy frame which is capable of holding and withstanding these loads from the clamp and payload.

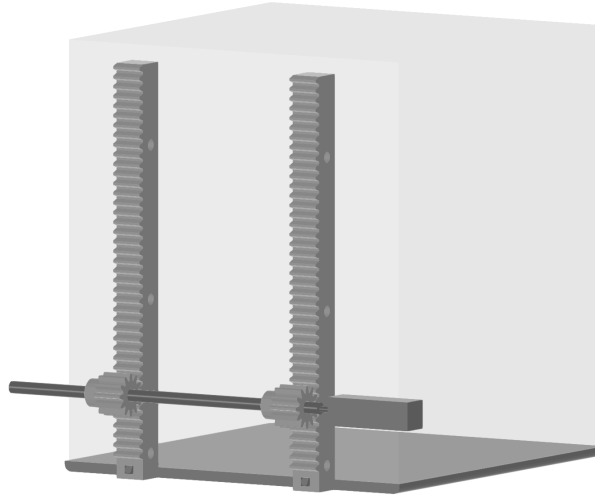


Figure 4. Clamp's rod, rails, gears and motor

The main idea is to have a heavy frame at the back end of the slot, the left-hand end of the rod and the motor would then be attached to the heavy frame (since the heavy frame is a circle which wraps around the fuselage as seen in figure 11 in appendix B. The motor will be connected to the heavy frames using multiple heavy-duty bolts, as these are able to withstand high values of stress before failing. As the rod will be spinning, it cannot therefore be connected directly to the heavy frame, instead bearings will be used. The end of the rod will be attached and welded to the inner opening of the bearing, the bearing will then be attached to the heavy frame, this way the rod will be able to spin while supporting the weight simultaneously. The outer diameter of the rod was found to be 6 mm [5], therefore a bearing with inner diameter of 6 mm and a maximum load capacity higher than 57 kg was found on the market [6] and used for the attachment.

5.2 Stringers and Spring Reels

After deciding on attaching the clamp using the rod and motor, structural analysis was done on the bottom plate, further details about this can be found in [1]. Figure 5 shows the bottom plate being loaded downwards to represent the payload, as it can be seen, there is a large displacement at the front end of the plate and the payload container would just slide off if this was the case. The two rectangular cavities that can be seen in the figure will be discussed in the following section.

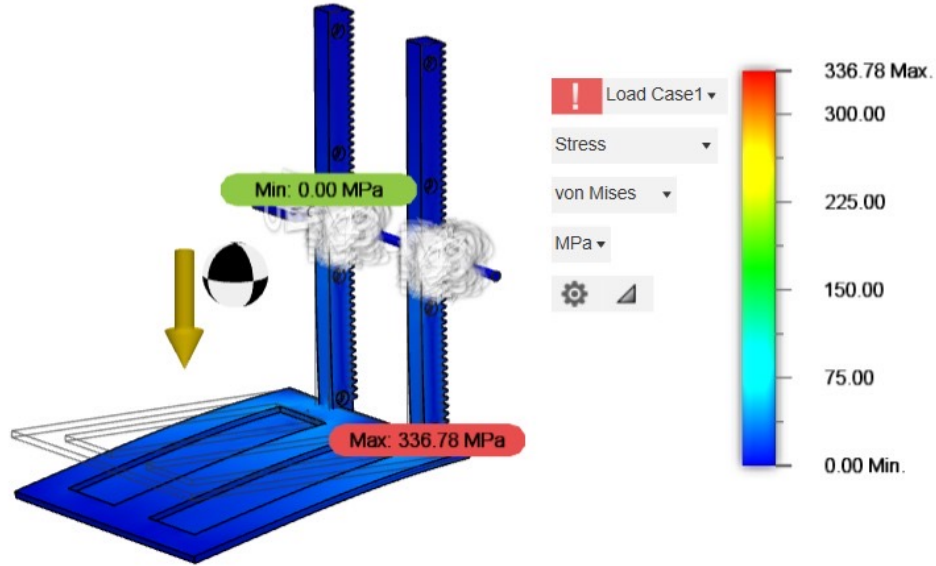


Figure 5. Structural analysis of bottom plate

After further analysis, the Structures team concluded that it would be better to attach the add-ons to two heavy frames instead of one, one heavy frame in front of the payload and one behind it. This change, along with the displacement problem mentioned above, required a modification to the original attachment method of the clamp to the fuselage. The plate now had to be supported from the front end and secured using two heavy frames.

One way of modification that was considered, was to add rails and gears to the front side of the clamp which would also be driven by another rod and a motor. The downside of this method is that the second motor would increase the total power usage and also add more weight, but more importantly, having rails on the front would only allow sliding the container in and out of the plate from the sides, which was not feasible since the Structures team decided to go with skids for the landing gear and this would therefore block the side access. Using a winch with cables attached to the front end was also considered, but the extra weight and power consumption from the winch were prohibitive.

Using retractable spring reels and stringers/cables was deemed the best option for this case. The cables would be attached to the plate and connected to the spring reels, the purpose of the spring reels is to allow the plate to be lowered or raised while still maintaining the tension in the cables to support the plate. In total, there will be four stringers and four spring reels, two on each side. One stringer will be attached to the front end of the plate and be connected to a spring reel which will be attached to the rear heavy frame, while the other stringer will be attached to the rear end of the plate and connected to a spring reel on the front heavy frame. This arrangement on each side would then form an 'X' shape with the stringers, improving stability and also preventing the container from sliding sideways.

Retractable spring reels were found on the market [7], these were chosen carefully to be able to carry 15 kg each. After some research and material selection for the stringers, it was found that Dyneema SK78 [8] with 3 mm diameter was a suitable choice for the stringers as they were cheap, lightweight and can withstand tension forces of up to 370 kg. A length of 1 m was chosen for each cable. The dimensions of the bottom plate were also increased slightly to make room for the cables

to attach. Figure 6 below and figure 12 in appendix C shows the final design of how the clamp will be attached to the two heavy frames, and it also shows that the design is structurally stable with no large displacements for the plate.

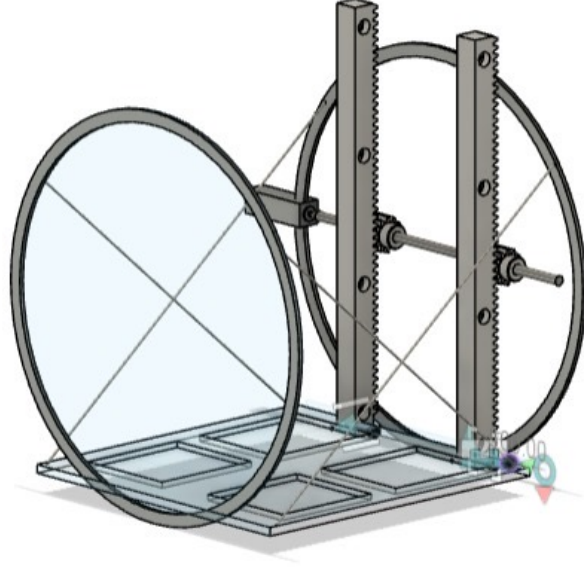


Figure 6. Final clamp attachment to fuselage design with stringers [1]

6 Aerodynamic Analysis

As it has been noticed in figure 5 above that there are two rectangular cavities in the bottom plate, this is explained in more detail in [1]. The main purpose was to lower the weight of the plate while maintaining structural performance by creating rectangular openings in the plate, this was found to give good structural performance. However, considering when the UAV delivers the payload and travels back without any container, the openings on the plate would not be covered and will allow air to flow inside the slot and create turbulence and separation, therefore reducing the aerodynamic efficiency and increasing drag. To solve this issue while still reducing weight, the underside of the plate was kept completely flat without any openings and the upper surface had areas with variable thicknesses which resulted in the cavities that are seen in the figures. Figure 13 in appendix D shows the final design of the bottom plate, the small square cavities have a thickness of 3 mm, while the rest of the plate has a thickness of 1 cm.

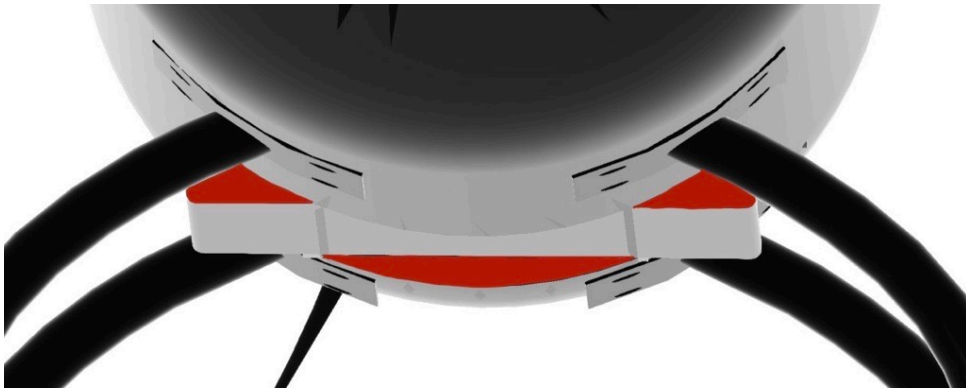


Figure 7. View of underside of fuselage with clamp and add-ons

Figure 7 above shows the final design of how the clamp and plate will hold the payload inside the

slot. As it may be noticed that the bottom plate does not curve to conserve the shape of the fuselage and reduce drag, this is mainly due to loading/unloading reasons, if the plate was to be curved from the bottom, this would then require the heavy payload to be lifted a few centimetres from the ground to slide it into the plate when the clamp is lowered on the ground, and this would just make loading and unloading much harder for the user. Having a curved underside of the plate would also make it more difficult for it to balance when it is left standing on the ground.

6.1 Miscellaneous Drag

Miscellaneous drag is the drag due to non-streamlined objects which are sticking out into the flow. The red-shaded areas in figure 7 are flat and are opposing the flow, therefore, these will definitely cause small amounts of drag. The estimation of the drag from the miscellaneous objects rely mostly upon test data. Equation (2) below is used to estimate the drag from the red-shaded areas in figure 7 above, this equation uses the total frontal area as a fraction of S_{ref} and multiplies it by 1.28 which is the drag coefficient due to a flat plate which is perpendicular to the flow (maximum drag).

$$C_D = 1.28 \frac{A_{frontal}}{S_{ref}} \quad (2)$$

The total red-shaded area was found to be $A_{frontal} = 242.1 \text{ cm}^2$, and S_{ref} was provided by the Aerodynamics team to be 4.235 m^2 . This then gave $C_D = 0.00732$, which is minimal and will not significantly affect the UAV's performance, according to the Aerodynamics team. As it may have also been noticed in the figure, the edges of the bottom plate have been curved slightly to help reduce the drag resulting from the plate sticking out.

6.2 Leakage Drag

Leakage drag is the further drag due to the small gaps and holes, as there is a tendency of an aircraft to 'inhale' through gaps and holes in the high-pressure zones (underside of aircraft). This tendency to 'inhale' contribute directly to drag. Since there will be small gaps between the plate and the slot, there will be extra leakage drag. It is not actually possible to calculate this extra leakage drag at this stage, therefore an estimate from [9] will be used which suggests that this extra drag is a percentage of about 1% of the total parasite drag (given by [10]).

As this is still the initial stage of the design and the extra drag is already minimal, further consideration will be given to reducing this additional drag in the next iterations or design stages, where further work will be done with the other teams to consider flattening out the fuselage from the bottom.

7 Communication Relay

It was mentioned in the introduction that there are two missions, one to deliver the 50 kg payload and the other long-endurance mission which will involve the UAV working as a communication relay system with a payload of 20 kg. The communication relay system would work by extending the signals in areas with no/weak coverage, more information about the purpose of this system and how it works can be found in [11]. The following subsections below explain how a new payload container and a new bottom plate were designed for the long-endurance mission.

7.1 Payload Container

A member [11] from the Systems and Propulsion team was working on the communication system, this member worked on designing the relay system and the equipment needed for it. The system will consist of a router and an antenna, the router can go anywhere in the UAV but the antenna has to be exposed to the outside. Since the router will be inside the payload container, extra holes had to be drilled into the new container to allow for the wiring to pass through and connect to the router.

Figure 8 below shows an opening of about 6 mm of diameter (given by Systems and Propulsion team) on the top of the container for wiring. The new container will still be made from the same material [1] due to its useful properties, and it will still have the same dimensions as the slot in the fuselage remains unchanged.

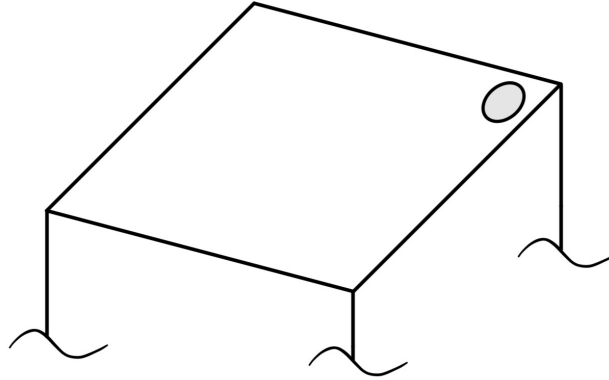


Figure 8. Top side view of the new payload container with opening for wiring

After obtaining the dimensions of the router from [11], it was noticed that the router will be much smaller than the payload container, and therefore, it will have to be secured in-place otherwise it will keep moving inside the container. Multiple approaches were considered to secure the router in-place, due to the container's material and other reasons, it was decided to go with a 3D printed structure to secure the router. Figure 9 below shows the router and antenna in orange, and the 3D printed structure is the thick black lines. The main idea is that the printed structure will extend to all four sides of the container to keep the router held and stop it from moving. The printed structure will be made from a lightweight but strong material able to withstand the forces. The final shape of the structure can also be changed depending on the weight restrictions and structural performance, this will be looked more into in the next design stages.

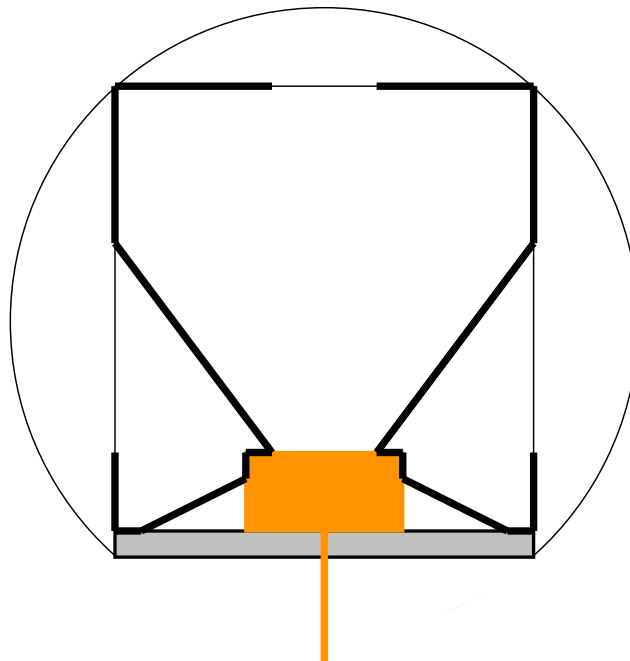


Figure 9. 3D printed structure to hold router (in orange)

7.2 Bottom Plate

Since the container will have an antenna protruding from it and the UAV will be used for various other markets (as mentioned in the following section), a new bottom plate had to be designed for each mission. The clamp is designed to have a swappable bottom plate. This will be achieved by attaching the plate to the rails using bolts, which can be easily screwed and unscrewed to swap the plate. More details about this can be found in [5].

The antenna will be protruding at an angle of 51.8° to the vertical according to [11], therefore the redesigned bottom plate will have an opening which is also at an angle to allow the antenna to extend outside. The plate will still be made from the same material and have the same dimensions and shape. Figure 14 in appendix E shows the plate with opening for the antenna.

8 Other Market Uses

Another objective was to enable this UAV to be used in additional markets, leveraging its modular add-ons to allow easy integration for various purposes with minimal modifications. The two main markets focused on were general monitoring (general land, infrastructure and crop monitoring), where the UAV will be equipped with the necessary equipment and sensors for monitoring when flying, and the other use was to spray pesticides on crops during flight.

For general monitoring, the UAV needs to be equipped with a camera. The idea is to have all the wiring and devices inside the slot and have the camera and sensors sticking out from the bottom. The Dji Zenmuse H20N [12] is a camera that was identified as a suitable option due to its high resolution and numerous features. This camera includes wide-angle and up to 32x zoom lenses, allowing for detailed identification of objects from long distances. It also features LiDAR scanners for detailed monitoring missions and night vision modes for use in low-light conditions. Additionally, the Zenmuse H20N [12] includes infrared and thermal scanners, which are valuable during these kinds of missions. The built-in gimbal ensures the camera remains steady regardless of the UAV's movements or vibrations. All these features make the Zenmuse H20N [12] ideal for monitoring applications.

The pesticide spraying would follow a similar idea where the pesticide itself will be stored in a container inside the slot and nozzles would then come out of the bottom plate to spray. Further details on the pesticide sprayer and the modified bottom plates for both the pesticide and monitoring uses can be found in [1].

9 Conclusion

It was seen how the design of the modular add-ons mechanism ensured that the add-ons can be easily swapped in under 10 minutes between the two missions. Using a Pugh matrix helped choose the most suitable attachment method (mechanical clamp) and with the slot configuration, they contributed to minimising aerodynamic drag while keeping the loading and unloading processes fast, reliable and requiring minimal effort. The material selection of the bottom plate along with the manufacturing method and the use of bearings, stringers and reels were essential processes in the design stage to ensure the plate would not fail and continue to hold the payload after many uses. Although the flat bottom plate along with the gaps between the plate and the fuselage, were responsible for extra drag ($C_D = 0.00732$), these values were small and does not have much of an effect on a UAV of this size. The use of this UAV with other markets also reflect the success of this modular add-ons design and demonstrate the easiness of how the UAV can be simply modified to change its purpose. Future stages of the design will involve working with other teams to try and reduce the extra drag even further, more work needs to be done on the 3D printed structure for the relay system, to ensure its structural performance and weight is acceptable.

References

- [1] Shreya Ramteke. HALO REACH: Structural Analysis and Design of the Modular-Add Ons. GDP report 02079050, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [2] Drs. Ribera Vicent, Armanini, Fasel, Shamsuddin, Seifikar, and Steiros. UAS for Disaster Relief (UDR24) Brief, 2024.
- [3] Dr. Emiliano Bilotti. *Materials 2 - Introduction and Materials Selection*. Department of Aeronautics, Imperial College London, 2022.
- [4] Ashby M. Materials Selection Software CES EduPack. GRANTA, 2022.
- [5] Adam T Rosenthal. HALO REACH: Add-on Mechanical System Design, Sizing and Calculations. GDP report 02027024, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [6] Stainless Steel Ball Bearings - 626-2Z [Internet]. URL https://www.accu.co.uk/rotary-bearings/605092-626-2Z-MTK?uk_google_shopping=1&c=3&gad_source=1&gclid=CjwKCAjwjQWzBhAqEiwAQmtgTxxSjW2Zd1EW0sudWGz9W03paQiJ7qbcW0hqxuyNjxtGWJiu3p3umBo&CeQ0QAvD_BwE. Accessed: 07-06-2024.
- [7] LiftingSafety Tool Spring Balancer [Internet]. URL <https://www.liftingsafety.co.uk/product/tool-spring-balancer-1273.html>. Accessed: 04-06-2024.
- [8] Dyneema SK78 Rope Breaking Strength [Internet]. URL <https://www.piratescave.co.uk/service/dyneema-sk78-rope-breaking-strength/>. Accessed: 04-06-2024.
- [9] Raymer D.P. *Aircraft Design: A Conceptual Approach, Sixth Edition*. Playa Del Rey: American Institute of Aeronautics and Astronautics, 2018.
- [10] Max F Lawlor. HALO REACH: Drag analysis methodology and validity. GDP report 02026199, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [11] Divij P Wankhade. HALO REACH: Navigation, Communications and Data Relay System. GDP report 02027906, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [12] Dji Enterprise Zenmuse H20N [Internet]. URL <https://enterprise.dji.com/zenmuse-h20n>. Accessed: 29-05-2024.

Appendix

A Clamp's Bottom Plate Covering Slot

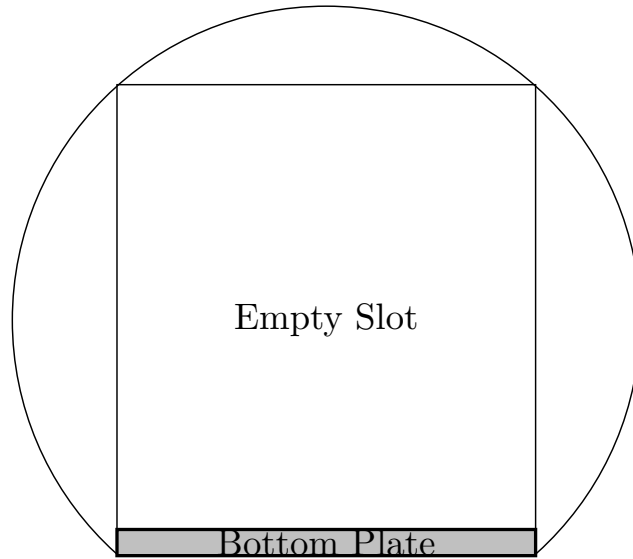


Figure 10. Bottom plate covering slot

B Heavy Frames

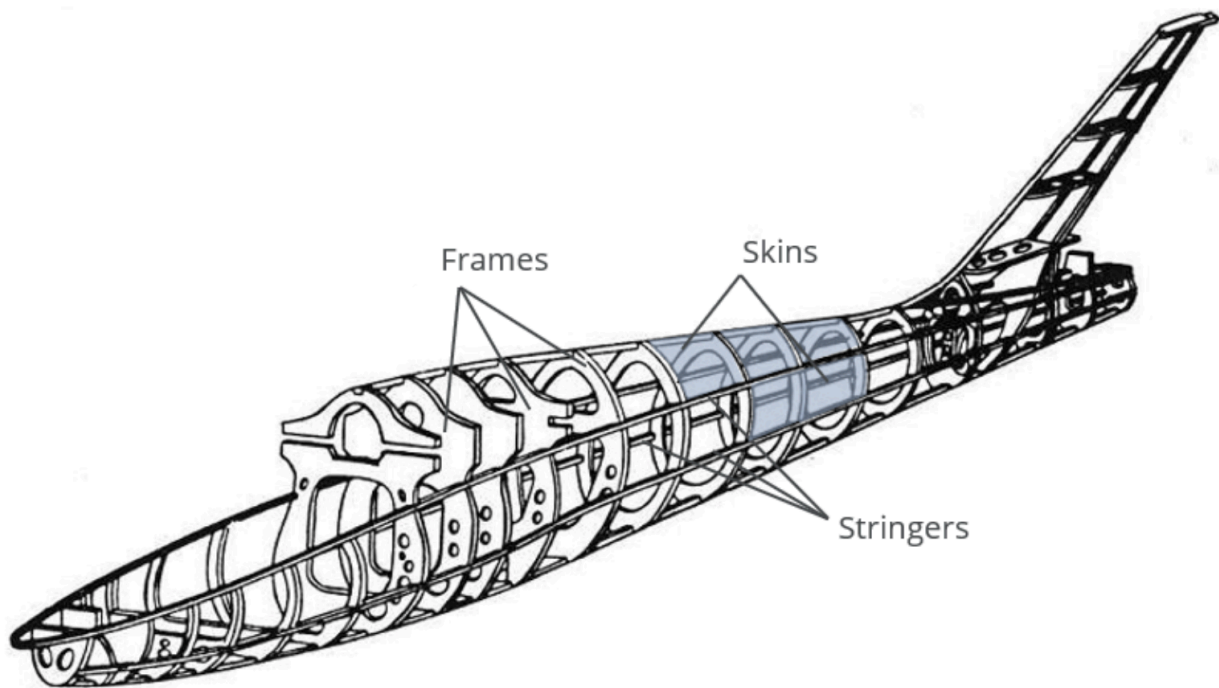


Figure 11. Image to show heavy frames on a fuselage

C Final Clamp Attachment To Fuselage Design

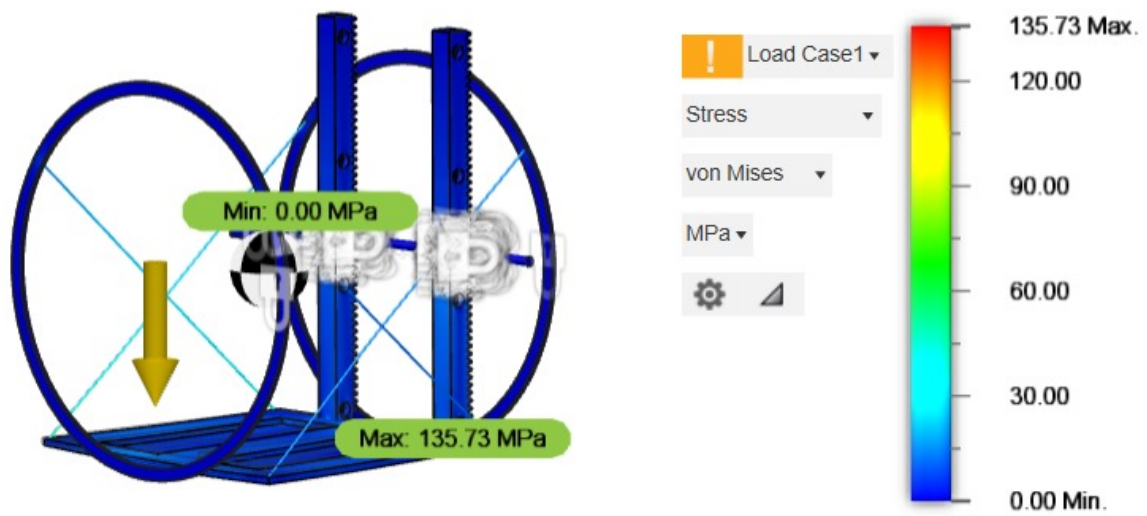


Figure 12. Structural analysis of final clamp attachment to fuselage design with stringers [1]

D Final Clamp's Bottom Plate Design

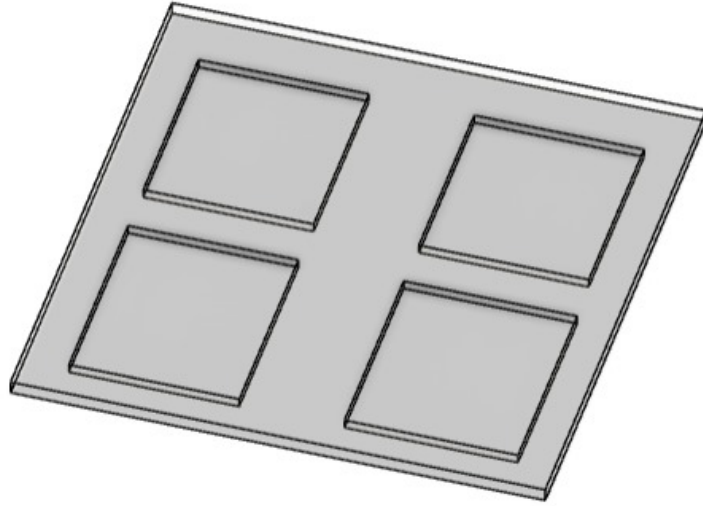


Figure 13. Final clamp's bottom plate design [1]

E Communication Relay Bottom Plate

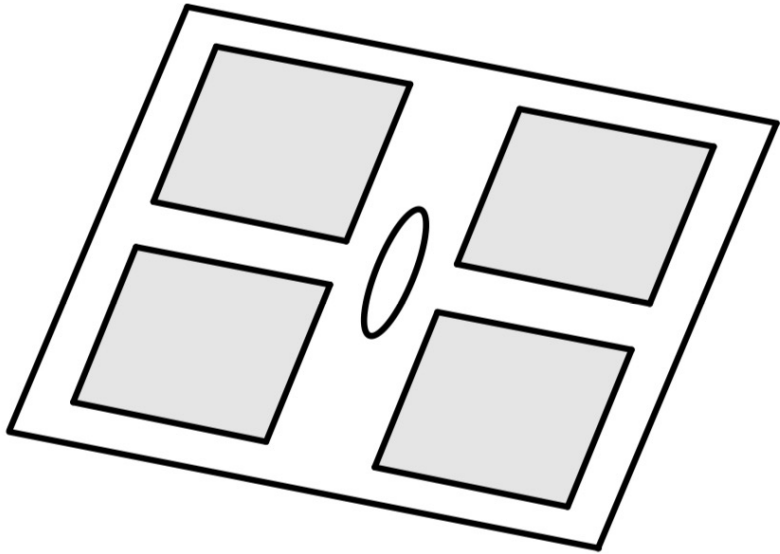


Figure 14. Communication relay bottom plate with antenna opening